Learning Rate Schedulers (Focus on Cosine Annealing)

A concise overview of how learning rate schedulers improve model training

Why Use Learning Rate Schedulers

- Fixed learning rates often lead to poor convergence or oscillation.
- Schedulers dynamically adjust the learning rate during training.
- Improves convergence speed, generalization, and training stability.

Common Learning Rate Schedulers

- Step Decay — reduces LR by a fixed factor every few epochs.
- Exponential Decay — LR decreases exponentially with time.
- Cosine Annealing — smoothly reduces LR following a cosine curve.
- Cyclic / OneCycle — periodically increases and decreases LR.

Cosine Annealing: Intuition

- Inspired by simulated annealing — 'cooling' the optimizer over time.
- Starts high, then gradually decreases LR following a cosine curve.
- Avoids abrupt LR changes and promotes smoother convergence.
- Encourages exploration early, fine-tuning later.

Cosine Annealing Formula

- Learning rate at epoch t:
- $LR(t) = \eta\_min + (\eta\_max - \eta\_min) / 2 \times (1 + \cos(\pi \times t / T\_max))$
- η_max — initial (maximum) learning rate
- η_min — final (minimum) learning rate
- T_max — number of iterations or epochs before restart

Cosine Annealing with Warm Restarts (SGDR)

- Introduced in 'SGDR: Stochastic Gradient Descent with Warm Restarts' (Loshchilov & Hutter, 2016).
- After T_max iterations, LR resets to η_max and decays again.
- Each restart helps the model escape local minima.
- PyTorch: torch.optim.lr_scheduler.CosineAnnealingLR or CosineAnnealingWarmRestarts.

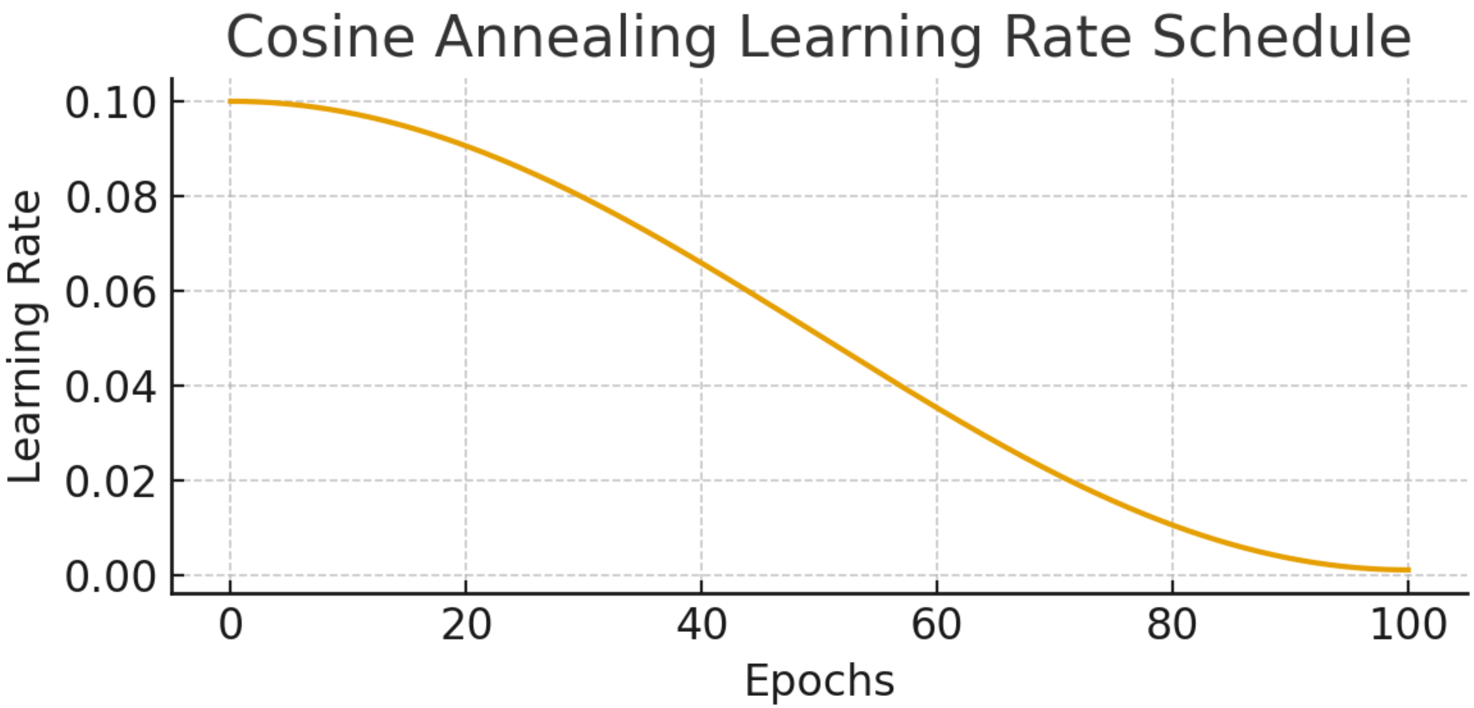
Practical Tips

- Choose η_max near your initial learning rate, η_min near 0.
- Use longer T_max for slower LR decay.
- Combine with warm-up for smoother training start.
- Monitor validation loss to fine-tune scheduler parameters.

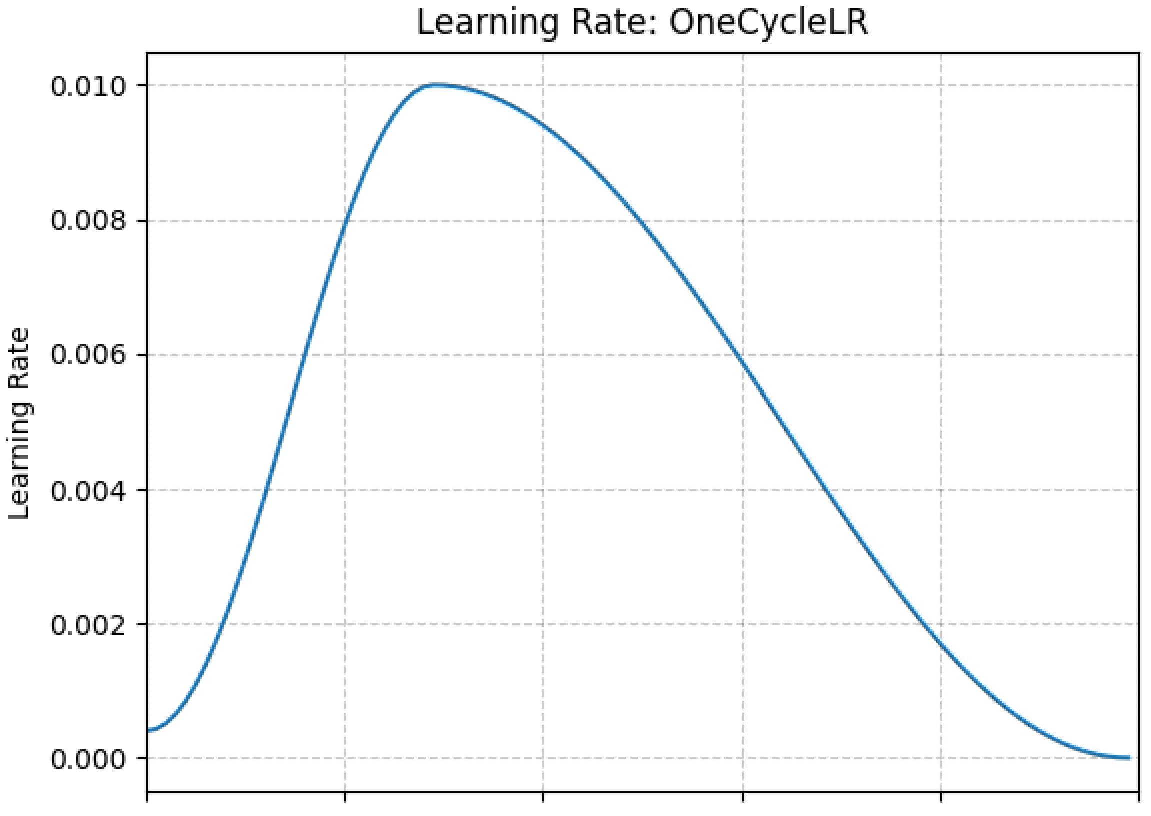
Summary

- Learning rate schedulers adapt training speed dynamically.
- Cosine annealing provides smooth, non-linear LR decay.
- Improves convergence and model generalization.
- Warm restarts enhance performance by reintroducing exploration.

Cosine Annealing — Visual Example



One Cycle Policy — Visual Example



Important: Do NOT Use Early Stopping with Schedulers

- Learning rate schedulers (e.g., Cosine Annealing, One Cycle) are designed to control the full training trajectory.
- Stopping early interrupts their intended schedule and prevents the model from reaching its optimal point.
- Instead of early stopping:Let the scheduler complete its planned cycle or epochs.
- - Use validation performance to adjust scheduler parameters, not to stop training abruptly.
- - Consider reducing max epochs if overfitting is a concern.